

EEE103/CSE161

Computer Programming

Lecture 03: Examples questions

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Switch and Case statements





2) Write a program that calculates the area of a circle.
Include prompts for input and display the results as output.

```
1  #include <stdio.h>
2  #include <math.h>
3
4  int main() {
5      float r;
6      printf("Enter the radius of the circle:\n");
7      scanf("%f", &r);
8      float area = 3.142*pow(r,2);
9      printf("The area of the circule is: %f\n", area);
10
11     return 0;
12 }
```

Enter the radius of the circle:

5

The area of the circule is: 78.550003



3) Write a program that checks if a given number is even or odd.

```
#include <stdio.h>
#include <math.h>

int main() {
    int n;
    printf("Enter a number: \n");
    scanf("%d", &n);

    if(n%2==0)
    {
        printf("The number is even\n");
    }

    else{
        printf("The number is odd\n");
    }
    return 0;
}
```

Enter a number:
2
The number is even

4) Write a program that finds the largest among three numbers.



```
#include <math.h>

int main() {
    int n1, n2, n3;
    printf("Enter three numbers: \n");
    scanf("%d %d %d", &n1, &n2, &n3);
    if(n1>n2)
    {
        if(n1>n3)
        {
            printf("Print %d is the greatest\n", n1);
        }
        else{
            printf("Print %d is the greatest\n", n3);
        }
    }
    else{
        if(n2>n3)
        {
            printf("Print %d is the greatest\n", n2);
        }
        else{
            printf("Print %d is the greatest\n", n3);
        }
    }
    return 0;
}
```

Enter three numbers:

9

2

132

Print 132 is the greatest



5) Write a program that takes numbers from the user and finds its average.

```
1  #include <stdio.h>
2
3  int main() {
4      int count, i;
5      float n;
6      float sum = 0;
7      printf("How many numbers do we you want to input?\n");
8      scanf("%d", &i);
9      printf("Enter the numbers.\n");
10     for(count=0; count<i; count++)
11     {
12
13         scanf("%f", &n);
14         sum = sum + n;
15     }
16
17     float ave = sum/i;
18     printf("The average value is: %f", ave);
19
20     return 0;
21 }
```

How many numbers do we you want to input?

3

Enter the numbers.

2.5

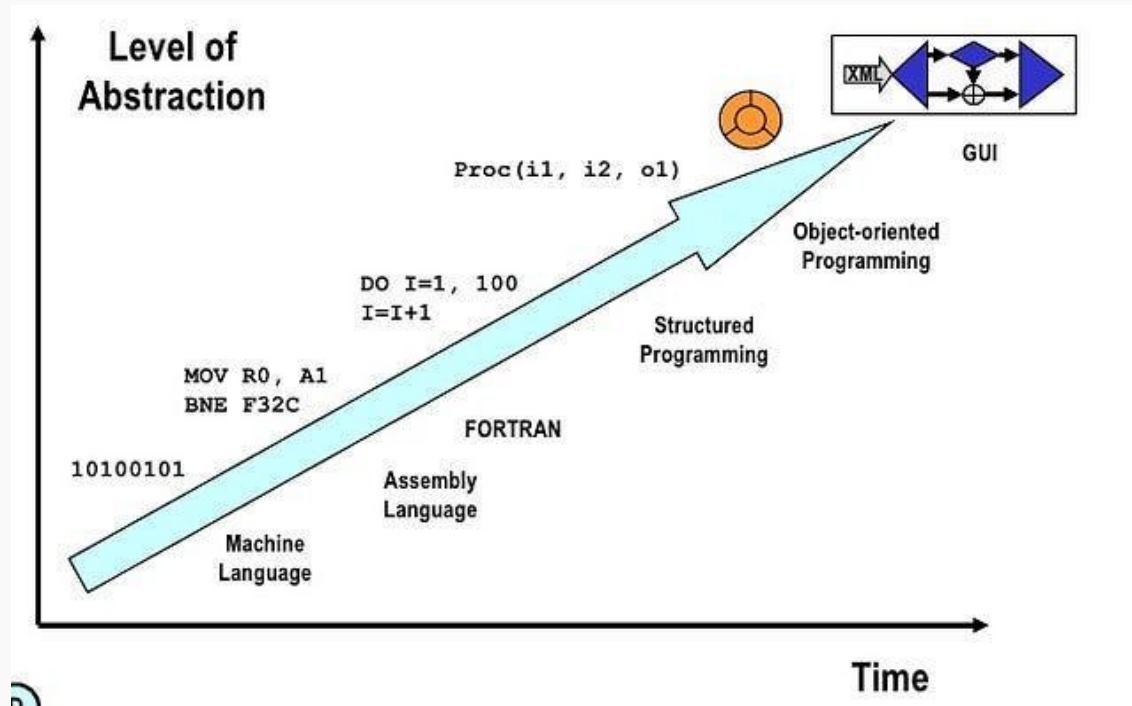
3.5

4.5

The average value is: 3.500000



Level of Abstraction





Getting Started With C

```
1
2  #include <stdio.h>
3
4  int main()
5  {
6      printf("Hello World");
7
8      return 0;
9  }
```

Return Type

Header File

Start of program

Library Function

Statement

End of program



Getting Started With C

```
2  #include <stdio.h>
3
4  int main()
5  {
6      printf("Hello World");
7
8      return 0;
9  }
```

- **include**, the # symbol indicates a preprocessor.
- This commands for something that has to be done compilation.
- The word **include** means to include the contents of the header file.



Getting Started With C

```
2  #include <stdio.h>
3
4  int main()
5  {
6      printf("Hello World");
7
8      return 0;
9  }
```

- `<stdio.h>` indicates name of the header file.
- The header file consists constants, functions and other declarations.
- For different purpose or different functions we need different library or header file.
- You may use 'help' or 'documentation' to find out which header file to call for a specific objective.



Getting Started With C

```
2  #include <stdio.h>
3
4  int main()
5  {
6      printf("Hello World");
7
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```

□ **<stdio.h>** indicates name of the header file.

□ The header file consists constants, functions and other declarations.

□ For different purpose or different functions we need different library or header file.

□ You may use 'help' or 'documentation' to find out which header file to call for a specific objective.

□ **stdio.h** means standard input output header file, which is typically necessary for taking inputs and printing them as output.



Getting Started With C

```
2  #include <stdio.h>
3
4  int main()
5  {
6      printf("Hello World");
7
8      return 0;
9  }
```

- In 2nd line 'main()' is a function, which is necessary to be called before every C program.
- Program starts from the first line of 'main()'.
- The term "int" represents the return type.



Getting Started With C

```
2  #include <stdio.h>
3
4  int main()
5  {
6      printf("Hello World");
7
8      return 0;
9  }
```

- The curly braces '{}' are like containers of the code.
- The code within an open and closed brace is known as a block.
- In case we miss either braces, the code will not be compiled and show compilation error, like "Compound Statement Missing".



Getting Started With C

```
2  #include <stdio.h>
3
4  int main()
5  {
6      printf("Hello World");
7
8      return 0;
9  }
```

- The function `printf()` is included in the header file `stdio.h`.
- Its function is to print any parameter.
- Every statement in C program must end with a semicolon ‘;’, otherwise it will generate error, like “Statement missing”.
- In the statement “`return 0`”, the term `return` indicates how the program ends/exits.
- In this case, if 0 is returned means there was no error in the code.
- Abnormal termination is typically indicated by non-zero return.



Some Programming Tools

- Compiler
- Standard Library
- IDE (Integrated Development Environment)
- Help files and Documentations



Compilers

- Match Syntax
- Finds errors
- Prepare object code to be translated to machine language
- Help files and Documentations

Standard Library

- Provide implementations of some basic and important functions
- Library functions must be used to increase efficiency and portability .



IDE - INTEGRATED DEVELOPMENT ENVIRONMENT

Helps to Write

- Use different color to highlight different type of code
- Sometimes shows hints

Helps to Compile

- Set environment variables
- Linking with libraries

Helps to Debug

- Execute step by step
- Use breaks
- Watch the values of variables

Helps to Run the code



Help file and documentations

- Provide details about
- Syntax
- Keywords
- Library functions
- Examples
- Etc.



Some important keywords

- C has some words that have special meaning for the compiler
- These words can not be used to **name variables, functions** etc.

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while



Variables

- ❑ Variables are like containers. They can hold values.
- ❑ The values can be of different types: Integer, Floating point numbers, Characters etc.
- ❑ Each variable takes up some memory space
- ❑ The values can be assigned, changed, read etc.
- ❑ Variables must be declared before using them



VARIABLE DECLARATION AND INITIALIZATION

- First write the keyword for data type
- Then write the name of the variable
- Example
- `int num=10;`
- `char c='a';`
- `int i, j, k;`
- `char esc='\\';`
- `float exp=3.2e-5;`
- `float sth=3.223;`



PRINT VARIABLES

```
1  #include <stdio.h>
2
3  int main() {
4
5      int x = 2;
6      double y = 2.04;
7      float z = 2.04;
8      printf("x = %d\ny = %0.2lf\nz = %0.4f", x, y, z);
9
10     return 0;
11 }
```

```
x = 2
y = 2.04
z = 2.0400
```

```
=== Code Execution Successful ===
```



Variable naming system

- Variables are case sensitive
 - Ccc, ccc, cCc & CCC are all distinct variables
- May comprise letters, digits and the underscore
- The first character in a variable name cannot be a digit
- Variable name cannot be same as any keyword
- No commas or blanks are allowed within a variable name

□

Data types



□ Basic datatypes of C-

- int (integer / whole number)
- float (4 bytes of memory space, 7 digits after the decimal point)
- double (8 bytes of memory space, 15 digits after the decimal point)
- char (character)
- void (**used in function prototype example no parameters**)

□

Allocated memory of the data types



Type	Size (bits)	Size (bytes)	Range
char	8	1	-128 to 127
unsigned char	8	1	0 to 255
int	16	2	-2^{15} to $2^{15}-1$
unsigned int	16	2	0 to $2^{16}-1$
short int	8	1	-128 to 127
unsigned short int	8	1	0 to 255
long int	32	4	-2^{31} to $2^{31}-1$
unsigned long int	32	4	0 to $2^{32}-1$
float	32	4	3.4E-38 to 3.4E+38
double	64	8	1.7E-308 to 1.7E+308
long double	80	10	3.4E-4932 to 1.1E+4932



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- `int i, j, k;`
- `char esc='\\';`
- `float exp=3.2e-5;`
- `float sth=3.223;`



Format specifier in C programming which act as a placeholder for an integer argument in a formatted input and output statement.

- **Integer: %d**
- **Character : %c**
- **Float : %f**
- **Double : %lf or %f**
- **Unsigned int: %u**
- **String : %s used in character arrays, will get back to it later**
- **In addition to digits, we have 3 special letters: h, l and L.**
 - **h**, used with integer numbers, indicates a **short** int (for example %hd) or a short unsigned int (for example %hu)
 - **l**, used with integer numbers, indicates a **long** int (for example %ld) or a long unsigned int (for example %lu).
 - **L**, used with floating point numbers, indicates a **long** double, for example %Lf

□

Arithmetic Operators





Thanks